

SOFTWARE REQUIREMENTS SPECIFICATION

Bridging Healthcare Gaps with AI

Project Code:

BHG-AI-(25-26)

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Submission Date:

October 20, 2025

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Document Information

Category	Information
Customer	Patients , healthcare providers (doctors).
Project	Bridging Healthcare Gaps with AI.
Document	Software Requirement Specifications
Document Version	1.0
Identifier	BHG-AI-(25-26)
Status	Draft
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Approver	Dr. Muhammad Ilyas
Issue Date	October 20, 2025
Document Location	Github
Distribution	Department of Computer Science

Definition of Terms, Acronyms and Abbreviations

Term	Description
API	Application Programming Interface
ASP	Active Server Pages
BP	Blood Pressure
CPU	Central Processing Unit
HTTPS	Hypertext Transfer Protocol Secure
JWT	JSON Web Tokens
ML	Machine Learning
MRI	Medical Resonance Imaging
NFR	Non-Functional Requirement
NLP	Natural Language Processing
RAM	Random Access Memory
SDK	Software Development Kit
SRS	Software Requirements Specification
URL	Uniform Resource Locator
PMC	Pakistan Medical Commission

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1. Introduction

The official agreement between the clients and developers is termed as the Software Requirements Specification (SRS) document, in which all the functions that a software needs to accomplish, are explained. Its main goal is to provide developers with a clear development roadmap by outlining exactly what users expect from the system [1]. In order to give everyone a comprehensive understanding of what is required, the document is divided into multiple sections. It lists the main features, what the system will do, how it will work with other components, and the constraints [2] that must be considered, when developing the software system discussed in this document.

1.1 Purpose of Document

The purpose of this document is to list all the functions, requirements and features that users need in our system. It tells what the system can do, what it wants to accomplish, how it works with other systems, what jobs it does, and how it solves problems. This document is written for stakeholders, customers, the team building the system, and the people who will test it.

1.2 Project Overview

Sync is a project that aims to make healthcare in Pakistan more accessible and less expensive. We will use this system to let users know better about the medical terms and help them better understand their health issues. The primary goal is to address our country's healthcare problems and offer everyone access to high-quality medical care. By creating an AI medical assistant that interprets medical reports and provides telemedicine doctor consultations for users, we hope to improve healthcare.

1.3 Scope

The software system being made is called Sync. It is being developed for patients who require both routine and urgent medical advice and for the physicians who want to conduct online consultations. This system's objective is to provide "automation support" for getting health advice, and connecting with the doctors online.

The system is accessible to everyone countrywide via a web browser. It runs on a cloud server, which means users can use it from any location with a smartphone or a laptop. This ensures that the system functions properly and that multiple users can access it simultaneously.

Any user is able to create an account on Sync. When registering, users can choose to join as either a patient or a doctor. Patients can interact with the AI health assistant and book appointments with doctors. Users can conduct video calls, voice calls, or text chat with doctors. The system also has medicine reminders that send notifications to users when it is time to take their medicine. The AI assistant analyzes the symptoms, advises users whether they need to see a doctor immediately or if they can manage their care at home.

Doctors are able to make their profiles with their specialty and location. They can accept appointment requests from patients, do online consultations through video, audio or chat, and see patient history. The system matches patients with the right specialist doctors based on their health problems.

The project has some limits. It does not connect with hospital computer systems or insurance companies. The system does not work with fitness watches or other wearable devices.

2. Overall System Description

Sync is a smart healthcare app. It uses Artificial Intelligence (AI) to help people with their health issues. The system makes it easier to understand the medical reports. Additionally, it speeds up the process of traditional doctor visits. Basic health checks are less expensive. It is fruitful to use in the areas far from the hospitals. The system analyzes the data with AI and connects patients with the right specialists.

2.1 User characteristics

This app will have three main groups of users. Our system will be developed for all types of people who need medical assistance.

2.1.1 Primary Users (Patients)

- These are the users who manage their health using the app on a daily basis.
- Men and women from both urban and rural areas are included in this.

2.1.1.1 Poor and Underserved People:

- People with limited financial resources, and do not afford expensive hospitals fee can utilize this system.
- They can get inexpensive basic medical assistance through our website.

2.1.1.2 Users in Remote Areas:

- People in villages or with slow internet speeds are able to use this system.
- They are capable to get remote help from doctors.

2.1.1.3 Elderly Patients:

- They require an extremely user-friendly app that is very simple to use.
- This enables their family members to assist in managing their healthcare.

2.1.2 Health Professionals

These users include doctors and other medical staff who help treat patients.

2.1.2.1 General and Specialist Doctors:

- Doctors who treat routine patients and those with certain illnesses can use the system.
- They are able to create their own account as a doctor.
- They can use the system to see patient history and give better treatment.

2.2 Operating environment

The operating environment makes sure that the system works properly and all users can access it easily. The detailed operating environment is given below.

2.2.1 Hardware

For optimal performance, the system needs the following user devices and hardware infrastructure.

2.2.1.1 Servers:

- Using cloud based hosting with auto-scaling features allows the system to handle multiple users at once.

2.2.1.2 User Devices:

- Desktop or laptop computers.
- Tablets.
- Smartphones (via mobile browser).

2.2.2 Software

To maximize the user reach, the system is available on different software platforms.

2.2.2.1 Web Application:

- The system works on all common web browsers like Google Chrome, Mozilla Firefox, Safari, and Microsoft Edge.
- Users can open the website on any browser and use the system.

2.2.3 Network

- The healthcare system requires internet connection to work.
- The system works on both mobile data (3G/4G/5G) and WiFi connection.

2.2.4 Security and Privacy

- All user data is protected with secure HTTPS connection.
- Patient information is kept confidential and secure according to privacy regulations.
- Only authorized users (patients, physicians, healthcare workers) can access the system.

2.3 System constraints

During the system's design, development and deployment phases, our healthcare system is subject to some constraints. These constraints define the boundaries within which the system will operate. These include:

2.3.1 Software Constraints

- The web platform works on modern browsers (Chrome, Firefox, Safari, Edge).
- Users must have internet connection to access the platform.

2.3.2 Hardware Constraints

- Host server (Netlify / Vercel) must have sufficient resources (CPU, RAM, disk space) to run the web application.
- Users need adequate devices (phones/computers with minimum 2GB RAM).
- Camera and microphone are required for video consultations.

2.3.3 Legal Constraints

- Pakistan healthcare regulations must be adhered to.
- Doctor verification is required.
- User consent for data collection is necessary.

2.3.4 User Constraints

- Users must have basic smartphone or computer operation skills to navigate the platform.

2.3.5 Environmental Constraints

- The frontend is deployed on Vercel/Netlify and the backend requires stable internet connectivity to remain accessible.
- The quality of a video consultation may be affected by various lighting and noise levels.

2.3.6 Off-the-shelf Components Constraints

- Groq API (LLM) and LangChain have rate limits and usage costs.
- Qdrant vector store has storage limits for the RAG knowledge base.
- NLP Models might not comprehend every medical term.
- Embedded SDK has usage limits and costs for video consultations.
- Third-party services may fail or become expensive.

3. External Interface Requirements

This section tells us what hardware, software and communication interfaces our system uses to work properly.

3.1 Hardware Interfaces

The system uses the following hardware components to support its core functionality.

3.1.1 Microphone and Speaker

- The system uses the smartphone or computer's microphone and speaker for video calls between doctors and patients.
- Echo cancellation helps make calls clear.

3.2 Communications Interfaces

3.2.1 Virtual Consultations

- Patients and doctors have video calls for their queries.
- The service works on slow internet connections too.
- All calls are encrypted for privacy and security.
- Audio quality adjusts automatically based on internet speed.

3.2.2 Network Communications

- All data sent between app and server uses HTTPS for security.
- User login information is protected with JSON Web Tokens (JWT).
- All patient data will remain encrypted.
- The communication works safely over the internet.

3.2.3 Message Formatting

- All AI communications is written in simple language.
- No complex medical terminology is used.

4. Functional Requirements

4.1 User Management

- 4.1.1 Users are able to register, login, and manage health profiles with medical history.
- 4.1.2 The system keeps the user sessions safe and offers password recovery feature.

4.2 AI Health Assistant

- 4.2.1 Using medical AI models, the system offers a chat interface for health-related queries.
- 4.2.2 Based on user's medical profile, the system gives personalized responses.
- 4.2.3 The system maintains conversation context and provides confidence scores for responses.

4.3 Diagnosis & Treatment Recommendations

- 4.3.1 After analyzing the symptoms, preliminary health advice is provided along with a recommendation to consult a doctor if needed.
- 4.3.2 The system classifies (Non-urgent/Moderate/Urgent/Emergency) and suggests treatments.

4.4 Appointment Booking

- 4.4.1 Appointment confirmation details are stored and accessible to the user.

4.5 Doctor Directory

- 4.5.1 A searchable doctor database with profiles and specialties is maintained by the system.
- 4.5.2 The system allows filtering by location, specialty, and availability.

4.6 Teleconsultation

- 4.6.1 During a consultation, the system shows the medical history of the patients to the doctor.

4.7 Medication & Health Tracking

- 4.7.1 The system stores medication reminder notifications in the database.

5. Non-Functional Requirements

5.1 Performance

- 5.1.1 AI response time is within a few seconds.
- 5.1.2 The system supports concurrent users, and pages load within a few seconds.

5.2 Reliability

5.2.1 The system maintains daily backups and a 95% availability rate.

5.2.2 The system handles errors gracefully and recovers automatically.

5.3 Security

5.3.1 The system uses encrypted passwords, JWT authentication, and HTTPS communication.

5.3.2 The system enforces role-based access control combined with session timeouts.

5.3.3 The system applies input validation and rate limiting.

5.4 Usability

5.4.1 The interface is intuitive which requires no training.

5.4.2 The design is responsive for smartphones, tablets, laptops or desktops.

5.4.3 The system displays clear error messages and real-time form validation.

5.5 Scalability

5.5.1 The modular architecture supports component-level scaling.

5.5.2 Database indexing and caching enhance performance.

5.6 Compliance

5.6.1 The system displays medical disclaimers stating that it is not replacement for professional care.

5.6.2 The system requires user consent for data collection with export and deletion options.

5.7 Compatibility

5.7.1 The system supports modern browsers including Chrome, Firefox, Safari, Edge.

5.7.2 The system functions on 3G networks and above.

6. Assumptions and Dependencies

Following are the assumptions and dependencies that have been made during the development of the system requirements:

6.1 Assumptions

6.1.1 Internet Connectivity

- Users (both patients and doctors) have access to stable internet connectivity.
- Internet infrastructure in Pakistan remains accessible and affordable for target users.

6.1.2 User Device Availability

- Patients have access to smartphones or computers with modern web browsers.
- Doctors have reliable devices for conducting tele consultations.

6.1.3 User Technical Literacy

- Users possess basic smartphone/computer operation skills.
- Patients can navigate simple web applications with minimal technical assistance.
- Doctors are familiar with telemedicine platforms.

6.1.4 Medical Data Availability

- Patients have access to their prior medical history and records.

6.1.5 Doctor Participation

- Licensed medical practitioners in Pakistan register on the platform.
- Doctors provide accurate credentials and specialty information.
- Doctors are available for scheduled consultations.

6.1.6 Legal and Regulatory Compliance

- The system conforms to data protection and healthcare regulations of Pakistan.
- Legally, medical disclaimers is adequate to shield the platform from responsibility.
- Doctor verification processes meet legal requirements in Pakistan.

6.1.7 AI Model Accuracy

- Pre-trained medical AI models are available and can be integrated.
- The accuracy thresholds can be met by AI models.
- AI models can be continuously improved if we collect user feedback.

6.1.8 Third-Party Service Availability

- Video calling and external APIs remain operational.
- Third-party service providers continue to use their current pricing models.
- API rate limits are sufficient for expected user load.

6.1.9 User Trust and Adoption

- Given the appropriate disclaimers, users trust AI-based health recommendations.
- Users are willing to share medical information on the platform.
- Telemedicine is accepted as a valid consultation method by the healthcare professionals.

6.2 Dependencies

This project depends on the following external factors and components:

6.2.1 Third-Party APIs and Services

- Video Calling Services: Whereby Embedded SDK for teleconsultation features.
- Groq API for powering the AI health assistant.
- Qdrant vector database for the RAG-based medical knowledge retrieval.

6.2.2 AI/ML Frameworks and Models

- Availability of medical NLP models that are pre-trained for symptom analysis.
- AI frameworks for diagnosis recommendations that are either open-source or licensed.
- Continuous availability of training data is required for model improvement.
- Medical knowledge databases for validating or verifying the AI responses.

6.2.3 Cloud Infrastructure

- Stable cloud infrastructure is required for system availability. The frontend is deployed on Vercel and the backend database is hosted via Supabase.
- Database Services are used to store user data, medical records, and system logs.
- Content Delivery Network (CDN) for quick content delivery across Pakistan.

6.2.4 Regulatory and Legal

- Compliance with Pakistan Medical Commission (PMC) guidelines.
- Adherence to Pakistan's data privacy laws.
- Compliance with changing telemedicine policies.

6.2.5 Network Infrastructure

- Stable telecommunications network in Pakistan that is 3G, 4G or 5G.
- Service quality is maintained by the Internet Service Providers (ISPs).

6.2.6 Development Tools and Technologies

- Development frameworks and libraries availability.
- Browser compatibility with latest web standards.

6.2.7 Financial Resources

- Funding for third-party service subscriptions and API usage.
- Budget for cloud infrastructure and hosting costs.
- Resources for ongoing maintenance and updates.

6.2.8 Human Resources

- Doctor data is seeded in the system; real doctor onboarding will require willing medical practitioners in future phases.
- Technical support team for system maintenance.
- Medical advisors for validating AI recommendations.

6.2.9 User Base Growth

- Gradual adoption by patients in urban and rural areas.
- Word-of-mouth promotion and user referrals.
- Marketing efforts to raise awareness about the platform.

7. References

Ref. No.	Document Title	Date of Release/ Publication	Document Source
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